Design parameters

V_{supply}	12 V	P_{OUT_total}	20 W
V_{LOAD}	90 V	f_{SW}	250 kHz
I_{LOAD}	very small		
V_{AUX}	12V		
I_{AUX}	20 mA		

Switching freq

$$R_T = \frac{2.21 \times 10^{10}}{250\,kHz} - 995 = 87.4\,k\Omega \rightarrow \boxed{R_T = 86.6\,k\Omega}$$

Transformer Selection

Want CCM operation

$$D = \frac{N_P/N_S \cdot |V_{LOAD}|}{V_{supply} + \frac{N_P}{N_S} \cdot |V_{LOAD}|} = \frac{\frac{1}{2} \cdot 90}{12 + \frac{1}{2} \cdot 90} = \underline{0.8}$$

Note: recommended $D < 0.5$ to reduce need for slope comp → will need slope comp

Chip provides programmable slope comp

Primary Winding Inductance

Letting $I_{LRR} = 0.7$

$$L_{M,calc} = \frac{V_{supply}^2 - V_{Lused}^2}{I_{LRR} \cdot f_{sw} \cdot P_{out} \cdot \left(\frac{N_s}{N_p} \cdot V_{supply} + |V_{cond}|\right)^2}$$

$= 25.7\ \mu H \longrightarrow$ HPH6-0325L at $26.3\ \mu H$

Ripple: $\Delta I_{LM} = \frac{V_{supply} \cdot D}{L_M \cdot f_{sw}} = 1.46\ A$

$$I_{C\,PEAK} = \frac{P}{V_{supply} \cdot D} + \frac{\Delta I_{LM}}{2} = 3.77\ A$$

Saturation $\wedge$

DCM Calcs

Use DCM for high voltage, low power, no slope comp

$V_{in} = 12V$ tolerance 10% → $V_{in\,min} = 10.8V$ $V_{in\,max} = 13.2V$

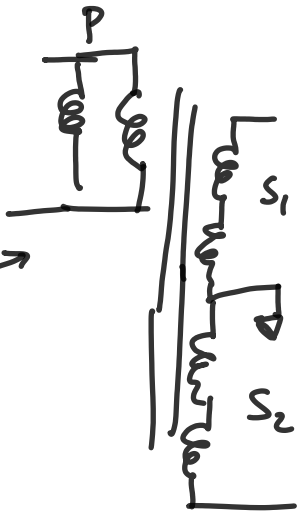
$V_{out} = 90V$; another secondary produces $-90V$

Haven't characterized transducer → assume $P = 1W$

Efficiency $\eta = 0.85$

$f_{sw} = 250\,kHz$

Turns ratio $\frac{N_s}{N_p} = \frac{2}{1} = 2$



Pick HP2-0216 transformer

$L_P = 10.6\,\mu H \pm 20\%$

$I_{PSAT} = 0.77\,A \cdot \frac{6}{1} = 4.6\,A$

$$R_T = \frac{2.21\times10^{10}}{250\times10^{3}} - 955 = 87.46\,k\Omega \rightarrow \text{use } 86.6\,k\Omega$$

Reflected voltage on primary: $V_r = \frac{90V}{2} = 45V$

Let $D_{max} = 0.8$ for flyback

On time: DT Reset time: $D_r T$

$D \cdot V_{in} = D_r V_r$ from volt-second balance

$D_{max} = D + D_r$

$$D = \frac{D_{max}}{1 + \frac{V_{in\,min}}{V_r}} = 0.65, \quad D_r = 0.15$$

$$L_{m,max} = \frac{(V_{in\,min} \cdot D)^2}{2 \cdot P_{in} \cdot f_{sw}} \; ; \quad P_{in} = \frac{P_{out}}{\eta} \approx 1.2\,W$$

$$= 82.5\ \mu H$$

Our $L_m = 10.6\ \mu H$ is fine

$$I_{pk} = \sqrt{\frac{2 P_{in}}{L_{m,min} f_{sw}}} = 1.05\,A \rightarrow \text{no saturation}$$

Actual D and Dr

$$D = \frac{L_m I_{pk} \cdot f_{sw}}{V_{in}} = 0.21 \qquad D_r = \frac{L_m I_{pk} \cdot f_{sw}}{V_r} = 0.049$$

Set I_{Lim} to be 30% higher than I_{pk} for current sense purposes

$$\rightarrow I_{Lim} = 1.37\,A \qquad R_S = \frac{V_{CLTH}}{I_{Lim}} = 75\,m\Omega$$

V_{CLTH} ranges 93–107 mV. Check that min V_{CLTH} still keeps I_{Lim} above I_{pk}: $\frac{93\,mV}{75\,m\Omega} = 1.24\,A$ ✓

No slope comp for DCM → $R_{SL} = 0$

CS filter

At 250 kHz $t_{ON(MIN)} = 150$ ns

Require $2 R_F \cdot C_F < t_{ON(MIN)}$ for current limit

Also $3 R_F C_F < \frac{1-D}{f_{SW}}$

→ $R_F C_F < 75$ ns → Pick $R_F = 30\,\Omega$ $C_F = 1.5$ nF

MOSFET Selection

Stress: $V_{in\,max} + V_r = 58.2$ V

Check FDC86244

$V_{DS\,max} = 150$ V

$Q_g = 6$ nC

For the flyback controller Q_g should satisfy $Q_g < \frac{35\,mA}{f_{SW}}$

$\frac{35\,mA}{250\,kHz} = 140$ nC → ok

Clamp

Estimate $L_{lk} = 0.05 \cdot L_m = 0.53$ µH

Chose clamp voltage of $1.5\,V_r = 67.5$ V

$P_{clamp} = \frac{1}{2} L_{lk} I_{pk}^2 f_{SW} \cdot \frac{V_C}{V_C - V_r} = 0.21$ W

$$R_{clamp} = \frac{V_c^2}{P_{clamp}} \approx 22k\Omega \qquad P_R \geq 0.42W$$

$$C_{clamp} \geq \frac{V_c}{\Delta V_c R_{clamp} f_{sw}} = 3.6nF \rightarrow \text{use } 4.7nF$$

↑ ripple ~ 5%

Diode: ESID for high reverse breakdown voltage and fast switching

Can be used for both clamp and output diodes